

# AJAY VENKATESH

Brooklyn, NY

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## Education

New York University, New York, NY

Sep 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

## Experience

### • Campus Mesh

Software Engineer

Jan 2026 – Present

New York, US

- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

### • Amazon

Software Development Engineer Intern

May 2025 – Aug 2025

Seattle, US

- Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access (**4+ roles**) using **OAuth, Midway SSO, and HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments**.
- Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; applied low-level design and design patterns, reduced backend execution time by **35%** and ensured reliability by unit and integration testing using **Mockito**.

### • New York University (Novel AI Technologies, Inc.)

Software Developer (On-Campus)

Aug 2024 – Present

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform (**React, Node.js, REST APIs**); deployed **Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue.
- Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

### • PricewaterhouseCoopers (PwC)

Senior Software Engineer

Aug 2021 – Aug 2024

Bengaluru, India

- Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
- Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
- Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
- Optimized data ingestion with **parallel processing** for **100K-row** from Excel; reduced execution time from **5 hours to 1 hour**.
- Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

## Projects

### • Advanced Expense Tracker (Java Desktop Application) | Java, JavaFX, JDBC, SQLite, JFreeChart

- Architected a modular **Java application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
- Implemented concurrent transaction handling and real-time cost splitting using **multithreading** and synchronized blocks to ensure thread-safe database updates.

### • Hospital Management System (Full-Stack Application) | React, Node.js, Express, MongoDB, SageMaker, Hugging Face

- Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
- Fine-tuned a **Hugging Face transformer-based model** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.

### • RoBERTa Fine-Tuning with Custom LoRA | Python, PyTorch, Hugging Face, AGNews

- Architected a **parameter-efficient fine-tuning (PEFT)** pipeline using a custom **Low-Rank Adaptation (LoRA)** implementation to optimize a **RoBERTa-base** model for news classification.
- Optimized model efficiency by freezing pretrained weights and injecting trainable rank-decomposition matrices, reducing trainable parameters to under **1M (934K)** while achieving **83.45%** accuracy on the private leaderboard.
- Conducted extensive hyperparameter sweeps for rank ( $r$ ) and alpha ( $\alpha$ ) values, utilizing **AdamW** and linear decay schedulers to stabilize training under strict memory and parameter constraints.

## Skills

- **Programming Languages:** C++, Java, Python, TypeScript, JavaScript (ES6), SQL
- **Web Technologies:** HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize
- **Database:** RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
- **Cloud:** AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs)
- **Tools:** Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Kubernetes, Power BI, QuickSight
- **AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL

## Certifications

AWS Cloud Practitioner

Azure Data Fundamentals

Stanford Machine Learning (Coursera)